# ERAM RAHMAN

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## EDUCATION

**Bachelor of Science in Computer Engineering — University of California San Diego** Expected Graduation: 2027
**Major:** Computer Engineering
**Relevant Coursework:** Data Structures and Algorithms, Software Engineering, Digital Systems, Computer Architecture, Operating Systems

## SKILLS

- **Programming Languages:** Python, C++, C, Assembly (ARM), HTML, CSS, Java, JavaScript, MATLAB, TypeScript
- **Web and Application Development:** React, Next.js, JavaFX, SQL, SQLite, Git/GitHub, REST APIs, Node.js, Three.js, Anthropic API, Agile Development, Flask
- **IT and Hardware:** PC Assembly, Hardware Troubleshooting, System Diagnostics

## EXPERIENCE

**GiosTech Software Engineering Internship | Python, Anthropic API, Three.js** March 2026 – June 2026
- Developed a full-stack web application using Three.js and a Python/Flask backend to render, manipulate, and interact with 3D human muscle models.
- Created an interactive AI chatbot using the Anthropic API to collect user-reported pain information, identify likely affected muscle groups, and recommend relevant at-home exercises.

**SLAM Researcher — ERL (Existential Robotics) | Python, MuJoCo, NoMaD** January 2026 – Present
- Developed a Python-based robotic arm simulation implementing Rapidly-exploring Random Tree (RRT) for collision-free path planning and obstacle avoidance.
- Contributed to a reinforcement learning research paper by developing a MuJoCo simulation environment and configuring the NoMaD model for vision-based waypoint prediction and autonomous navigation.
- Co-developed a robotics planning workshop for UC San Diego and San Diego State University, preparing technical presentations on robot perception and motion planning.

**PC Technician and Owner** February 2019 – Present
- Diagnosed and resolved hardware, software, and operating system issues for 100+ clients, including component failures, driver conflicts, and system instability.
- Performed PC assembly, hardware upgrades, and system optimization to improve performance, reliability, and overall system efficiency.
- Reduced client downtime by an estimated 30–50% through troubleshooting, component replacement, and maintenance.

## PROJECTS

**PC Recommendation Website | React, Next.js**
- Developed a full-stack web application using Next.js (React), achieving perfect Google Lighthouse scores (100/100/100).
- Designed a recommendation algorithm that evaluates user survey responses to identify optimal PC and laptop options based on budget, workload, and hardware performance requirements.
- Integrated external APIs (Canopy API) to fetch product data and implemented server-side filtering on around 20 products per request based on user survey inputs.

**Voice Controlled Navigation | React, CSS, HTML**
- Built a web-based navigation application using Mapbox API with real-time GPS tracking and routing, achieving 150–170 ms API response times and a Google Lighthouse performance score of 100.
- Implemented voice command functionality for hands-free navigation, reducing manual interaction by 25–35% and improving user safety.
- Developed dynamic route recalculation and multi-stop navigation, thus improving routing efficiency.

**Student Clock Application | Java, JavaFX, SQLite**
- Developed a desktop productivity application using JavaFX and SQLite, integrating task management, time tracking, and persistent data storage.
- Implemented a priority-based scheduling algorithm to optimize task allocation and improve study time efficiency by approximately 50%.
- Developed clock-in/clock-out functionality, dynamic schedule rendering, and input validation to support accurate time tracking and reliable user interactions.